## Physical Diffusion

Assume that this process can be model with a Gaussian distribution

## Flow Matching

- Forward diffusion simulation free in a flow matching context
- We have a closed form for getting from clean image to noise image at time step $t$

## High level diagram

- Reduces compute overhead

## VAE Architecture

- Encoder/decoder are symmetric in the spatial up and down sampling

## Residual Blocks

- This approach keeps the gradient clean, whereas post activation that was common in early residual networks has a noisier gradient, making it harder to learn from

## Attention Blocks

- The speedup of flash attention 2 is in memory management

## Spatial Resizing

- Bilinear and bicubic interpolation introduces blurring that needs to be discarded by the VAE
- The simpler approach produces less complex artifacts

## FSQ Layer

- No need for loss adjustments to utilise the whole codebook

## Why $2\sigma(1.6) - 1$?

- Quite clsoe to uniform distribution, making codebook utilisation rise faster

## VAE Summary

- All of this is to make the DiT's job easier

## Training the VAE: Loss

- L1 handles structure
- Adversarial handles high frequency details like fur and skin

## Training the VAE: Warmup and Cooldown

- The discriminator can't be too strong too early

## What's been skipped?

- Spectral norm keeps the discriminator's gradients from exploding and feeding back to the VAE

## Attention

- RoPE is virtually required because the transform is position blind, whereas RoPE gives an ordering on neighbouring patches

## Adaptive Layer Norm

- Initialised to 0, which allows